

Multiple Choice (10 marks)

1. Which of the following is true of interrupts?
 - (a) They are generated when memory cycles are "stolen".
 - (b) They are used in place of data channels.
 - (c) They can indicate completion of an I/O operation.
 - (d) They cannot be generated by arithmetic operations.
2. How is IP address spoofing detected?
 - (a) Installing and configuring a IDS that can read the IP header
 - (b) Comparing the TTL values of the actual and spoofed addresses
 - (c) Implementing a firewall to the network
 - (d) Identify all TCP sessions that are initiated but does not complete successfully
3. You are given a message (m) and its OTP encryption (c). Can you compute the OTP key from m and c ?
 - (a) No, I cannot compute the key.
 - (b) Yes, the key is $k = m \text{ xor } c$.
 - (c) I can only compute half the bits of the key.
 - (d) Yes, the key is $k = m \text{ xor } m$.
4. A hash function guarantees the integrity of a message. It guarantees that the message has not be
 - (a) Replaced
 - (b) Overview
 - (c) Changed
 - (d) Violated
5. What are the port states determined by Nmap?
 - (a) Active, inactive, standby
 - (b) Open, half-open, closed
 - (c) Open, filtered, unfiltered
 - (d) Active, closed, unused

True / False (10 marks)

Answer True or False

6. True or False: Snort is a widely used tool for network intrusion detection and real-time traffic analysis.
7. True or False: CFB is a commonly used block cipher operating mode for encryption in electronic communications.
8. True or False: Snort is a popular tool for network analysis in multiprotocol diverse networks.
9. True or False: ARPANET is a modern network used for anonymous file transfers similar to Freenet.
10. True or False: In a complete K-ary tree of depth N, the ratio of nonterminal nodes to total nodes is approximately $(K-1)/K$.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to search a literals string in a string and also find the location within the original string where the pattern occurs by using regex.
12. Write a function to find t-nth term of geometric series.

End of Paper